

Strong computer-science and algorithmic problem-solving background with hands-on C++ and Python, compiler engineering, and independent research-oriented projects. I work in a hypothesis → experiment → validation loop: exact oracles, randomized/metamorphic tests, Python harnesses, measurable benchmarks, and reproducible counterexamples. I have implemented LLVM optimizations, graph algorithms, and program-analysis tools; the same workflow transfers directly to data-driven quantitative research.

Algorithms & competitions

HSE Vysshaya Proba prize-winner; overall winner of the MEPhI Junior competition in 2025 and 2026; Grand Prize at the Baltic science and engineering competition

Hypothesis → experiment

Exact oracles, randomized/metamorphic testing, adversarial counterexamples, reproducible harnesses, and benchmarks

C++ & Python

LLVM/C++23 optimization work, program analysis, and Python tooling for automated hypothesis validation

EXPERIENCE

- July 1 — August 31, 2026

MCST — compiler engineering intern
 - Implemented an LLVM loop-invariant code-motion pass in C++; reasoned about loop invariance, side effects, memory access, speculative safety, and loop structure.
 - Built a Python verification harness comparing execution behavior before and after optimization and checking expected transformation / no-transformation cases.
- 2026

PS-form Memory Dependence Analyzer
 - Ranked 1st of 49: the only 5.0/5.0 result and 104/104 official tests.
 - Built an exact oracle for small domains, metamorphic/randomized checks, and reproducible wrong-answer/time-limit counterexamples.
- 2024 — present

UniversalToolchain / Wist2 — creator and primary developer
 - Independently designed and developed a complex compiler/language system with intermediate representations, SSA/optimizations, and interpreter/CIL backends.
 - Use benchmarking, differential/metamorphic verification, and reproducible checks to compare implementations and configurations.

PROJECTS

- PS-form Memory Dependence Analyzer**

Ranked 1st of 49 with 104/104 tests, with reproducible wrong-answer and time-limit counterexamples.

UniversalToolchain / Wist2

1,465/1,465 tests with zero failures across nine packages and clean-consumer projects, plus interpreter/CIL parity and reproducible configuration checks.

x86-64 Codegen & Register Allocation Lab

A measurable pipeline with spill/reload metrics, code size, and reproducible result comparison.

SKILLS

- Programming**

C++23, Python, C17, C#, Rust

Algorithms & CS

data structures, graph algorithms, program analysis, CFG/SSA, compiler optimizations

Experiments & verification

hypothesis formulation and validation, exact oracles, differential/metamorphic testing, randomized testing, benchmarks, counterexample analysis

Research tooling

LLVM, Linux, Git, CMake, Python harnesses, AI/LLM-assisted research with independent verification

EDUCATION

Software Engineering student at HSE University

RECOGNITION

HSE Vysshaya Proba prize-winner; overall winner of the MEPhI Junior competition in 2025 and 2026; first-degree diploma and Grand Prize at the Baltic science and engineering competition.

Moscow / remote